

Stroke Amnesia: Why Language Models Still Need a Canvas

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Abstract

Large language models can describe drawings with ease and emit SVG, Canvas, or TikZ code that produces recognizable images. Yet I argue that these abilities do not add up to true drawing competence. Inky is an open-source, JavaScript-first drawing and animation system designed to help coding agents turn storyboard references into hand-drawn, code-rendered 2D animations. It combines a small canvas interface with repeatable tools whose results can be checked: brushes that imitate physical media, traced reference outlines, style cues, a browser view for checking the result, and editable drawing parts. I use Inky as a case study to explore why coding agents struggle with drawing. I identify consistent failure patterns: *stroke amnesia*, *compositional collapse*, and *patch reflex*. I show how written notes outside the chat, positions taken from references, immediate visual feedback, and repairs that fix the cause of an error help outputs stay consistent. I contend that the missing ingredient is not better pixel generation, but a persistent drawing surface that can track object identity, which parts belong to which objects, and where objects touch across generation and revision. This paper centers on the design philosophy behind this drawing loop and the structural challenges it exposes for LLM-assisted drawing.

Keywords: large language models, drawing, coding agents, canvas, visual reasoning, repair.

Introduction

A model that can describe a hand is not yet a model that can keep the wrist attached. This paper examines that gap. Modern language models can act through canvases, SVG, browsers, and visual feedback. They can write code that draws an image and can often describe what is wrong with that image. What they struggle to maintain is the drawing’s structure: which part belongs to which object, where objects touch, and what must stay consistent after a revision.

I ground the argument in INKY, a canvas drawing and animation system for coding agents. Given a storyboard, INKY gives the agent a stable workspace: references can be viewed and measured, important outlines can be traced, style can be translated into brush choices, and visible marks can be redrawn with clear settings. The system’s rules come from practical failures observed during use, including floating limbs, disconnected clothing, speech bubbles pointing at the wrong speaker, guide marks turning into final objects, and texture being used to hide unclear construction.

This paper makes three contributions.

1. A vocabulary for common failures in LLM-assisted drawing, grounded in a working animation system rather than synthetic prompts, including three named phenomena: *stroke amnesia*, *compositional collapse*, and *patch reflex*.
2. A design philosophy for storyboard-to-canvas animation that treats drawing as a process that can be checked rather than one-step image generation.
3. A practical account of how positions taken from references, explicit brushes, editable drawing parts, and cause-based repair reduce common drawing failures without hiding them inside a system whose choices cannot be inspected.

My aim is constructive. The paper distinguishes recognizable Canvas or SVG output from robust drawing competence, while focusing on how the drawing is made, checked, and repaired.

Background and Scope

Definition of “drawing”

In this paper, drawing means placing marks on a 2D canvas so that objects, bodies, text, motion, and texture are readable without the source prompt. In INKY, this includes storyboard redraws, comic frames, captions, speech bubbles, hand-drawn texture, and frame-by-frame animation. The scope excludes photorealistic text-to-image diffusion, 3D modeling, music notation, and handwriting as a separate recognition problem. The focus is the readability of 2D illustration made through code.

Definition of failure

The failure described here is not mainly aesthetic. It is structural. Parts detach, proportions drift, intended objects become blobs, and repairs are made with patches instead of rebuilt shapes. A model can write a correct paragraph describing a person holding a mug and still draw a hand floating beside the cup.

Why now

This problem is now easier to see because models can inspect their own output and revise code in a loop. The question is no longer only *can the model emit drawing code?* It is also *can the model repair drawing structure when shown that it is wrong?*

How INKY Works

Overview

INKY renders 2D animations from user-provided storyboard references using a coding agent. The agent writes and revises the drawing code, while INKY provides the canvas, preview, reference tools, and export controls. The system does not ask the model to produce a finished image in one step. It asks the model to make a drawing in a workspace where marks can be checked and revised.

Drawing process

The process is deliberately simple: interpret the reference, take positions from it where placement matters, choose a brush style, redraw the scene with canvas code, inspect the

result, and fix the cause of any visible error. The agent remains responsible for style, timing, and composition, but it works inside a workspace where important marks have a visible reason in the code. This is the central design choice in INKY: the system does not attempt to make drawing automatic; it makes drawing easier to check.

Why Canvas

Canvas is repeatable, browser-native, and easy to render frame by frame. INKY avoids pasting source images or relying on generated image output because the goal is controlled drawing. Every visible mark should come from a line of code or a setting that can be changed.

Design principles

Five principles guide the system. First, creative decisions remain with the agent; INKY supplies drawing and review tools, not finished templates. Second, rendering is repeatable; brush texture and jitter use the same random settings so the same frame can be redrawn without accidental changes. Third, reference images are used as guides: important outlines, lettering, props, and poses can be measured before they are redrawn. Fourth, style choices are explicit. Instead of asking for something vaguely “hand-drawn,” the drawing code names brush size, roughness, texture, transparency, and random setting. Fifth, important parts can stay editable, so later changes move the object rather than paint over pixels.

Keeping context outside the chat

The agent’s instructions are stored in project files, not only in the conversation. Requirements, design rules, reference notes, notes about drawing parts, and review notes persist across revisions. This gives the agent a form of memory outside the chat.

What the tools do

The tools have three simple jobs. They give the agent visual evidence: references can be overlaid, outlines can be traced, and style references can suggest colors and brush settings. They make marks explicit: a brush stroke has a color, size, texture, transparency, and random setting. They also make repair possible: important parts can be kept as editable drawing parts rather than anonymous pixels. The tools do not replace judgment. They make the drawing process easier to inspect.

Three Named Failure Modes

In my work with INKY, three failure modes recur often enough to need names. They are not mutually exclusive; a single broken frame can show all three. Naming them makes prompts, review notes, and repair decisions more precise.

Stroke amnesia

A model can draw a mark and later lose track of what that mark was for. A guide arrow can become a wire. A construction line can become decoration. A speech-bubble tail can drift toward the wrong speaker. I call this *stroke amnesia*. The issue is not that the model cannot explain the

difference between an arrow and a wire. The issue is that the drawing has no durable record of which marks are planning marks and which marks belong to the final scene.

Compositional collapse

The model can draw a cat. It can draw a kettle. Asked to draw a cat to the left of a kettle on a table, with the cat watching steam rise, relations begin to fail: the cat sits on top of the kettle, the kettle floats above the table, or the steam emerges from the cat. Composition breaks when several relations must hold at once: a character talks while holding papers, a bubble must avoid the face, a prop must remain on the desk, and the same object must stay visible across frames.

I call this *compositional collapse*: relations fail as the number of simultaneous constraints rises. Visual feedback helps, but it does not by itself teach the model which objects own which parts.

Patch reflex

Shown a defect, the model often covers the visible problem instead of fixing the broken construction. It may add a white patch, an eraser stroke, a transparency trick, or extra texture. These fixes can work in a single still image, but they usually fail when the scene moves or the same object appears again. I call this *patch reflex*. The repair rule in INKY is simple: fix the cause and redraw the broken shape or layer. Do not cover it.

Why these three matter

Stroke amnesia is a failure of identity. Compositional collapse is a failure of relation. Patch reflex is a failure of repair. Together, they show why language fluency and code fluency are not enough. The model needs a stable way to remember what each mark belongs to and why it was drawn.

How INKY Helps

INKY responds to these failures by making the drawing process more visible. The following practices give the agent something stable to reason with.

Reading the reference

Before detailed rendering, the agent reads the reference as a set of important relationships: where body parts connect, where objects touch, which shapes are easy to misread, and which marks belong only to planning. This reference reading is not a copied layer. It is a guide for what must remain true when the scene is redrawn.

Drawing from a reference

When a reference image is provided, the agent follows a simple order: show the reference, trace important outlines, study the style, choose brush settings, redraw with canvas strokes, compare the result, and hide the reference before export. The reference is evidence, not a hidden bitmap. Traced outlines provide positions; studying the style provides color and texture hints; the final frame is still redrawn in code. This order matters because it reduces guessing without removing authorship.

Keeping parts connected

The most important response is to treat visual marks as owned parts of a scene rather than anonymous pixels. A hand belongs to an arm; a speech bubble belongs to a speaker; a construction arrow belongs to the planning layer rather than the final drawing. This makes repair more direct: if a hand is detached, the arm or hand placement should be corrected, not covered. White overlays, transparency tricks, extra texture, and rough eraser edges are rejected because they can make a single frame look acceptable while leaving the underlying scene structure broken.

Project files as memory

INKY stores drawing context in project files: requirements, reference notes, lists of important parts, frame notes, and review notes. The agent re-reads the relevant file instead of relying entirely on conversation memory. These records give the drawing a form of memory outside the model.

Related Work

This framing draws on three strands. *Text-to-image evaluation* grades alignment of generated images to prompts and is the closest neighbor in spirit, although it operates on generated pixels rather than process. *Vector and symbolic visual representation* evaluates LLMs on SVG and related code-based images, and is directly relevant to the claim that emitting valid markup is not the same as drawing. *Sketch generation and sketch datasets* study sketching as a generative act, often treating strokes as steps in a drawing. INKY adopts that view but pairs it with coding-agent repair, which most prior sketch work does not address.

This paper distinguishes fluent SVG or Canvas generation from reliable drawing. The argument builds on work that treats drawing as a sequence of decisions rather than only a final image.

Limitations

This paper reports observations from a single working system. The phenomena described here are recurrent in that setting, but I do not generalize them across all models or tasks. The named failures are a working vocabulary for interpreting problems observed in LLM-assisted drawing. The scope excludes photorealistic image generation and 3D modeling; readers working in those areas should treat the claims as specific to code-rendered 2D illustration.

Conclusion

A model that can describe a hand is not yet a model that can keep the wrist attached. I have argued that current LLMs lack a persistent drawing surface: a way to remember object identity, which parts belong to which objects, where objects touch, and why marks were drawn. INKY works around this gap with project files, visual feedback, and a rule that visible errors should be fixed at their cause. This is useful because it turns drawing into a process that can be checked. The central lesson is that final rendered images are not enough: the way marks are made, revised, and repaired is part of the drawing.

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